

Report

Assignment 3

Data Analytics

E0-259

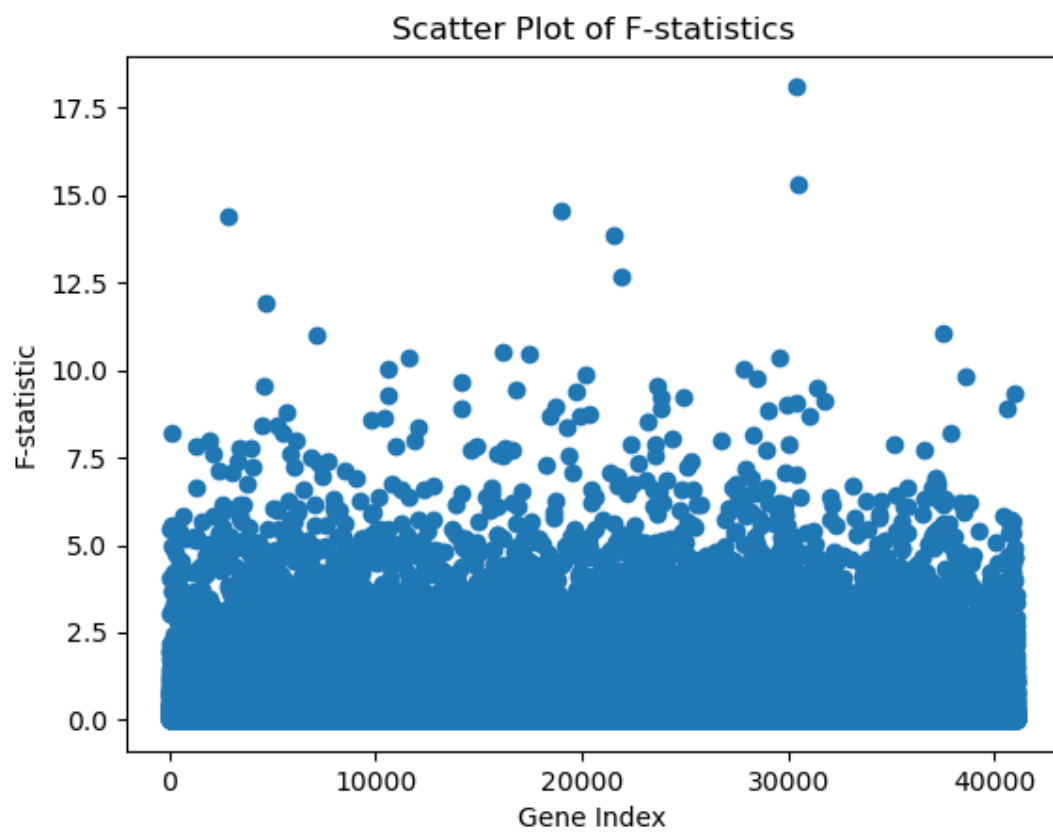
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Output:

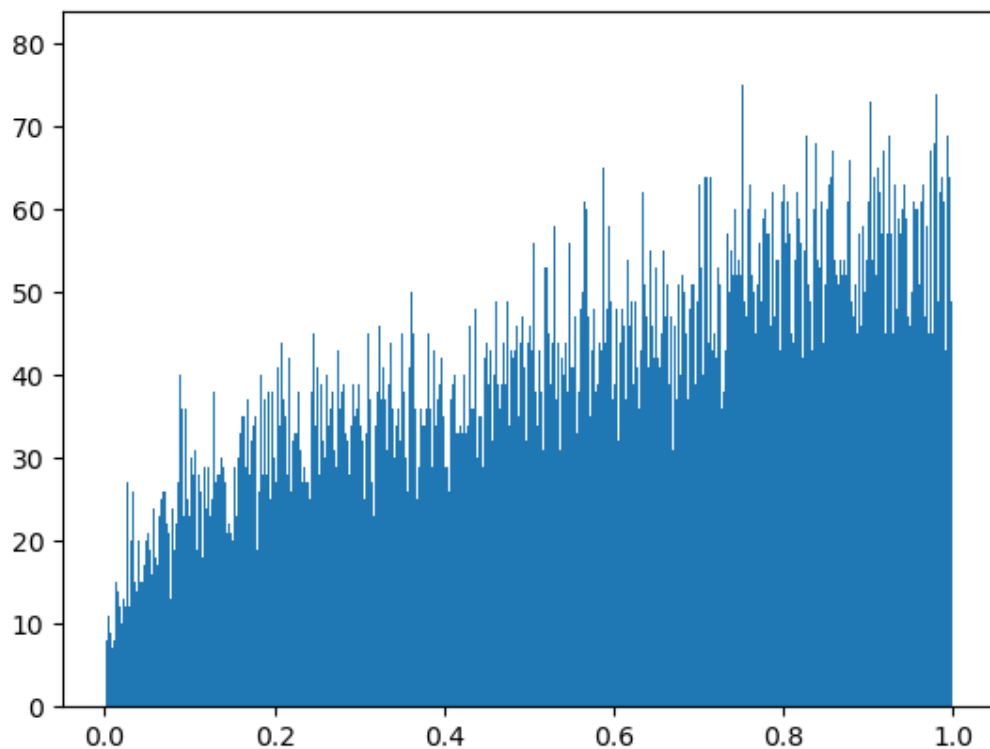
The number of significant or genes of concerned are 702.

The threshold for f-statistics was set at 4.08 by looking at f-table.

The histogram of F-statistics



The histogram plot of p-values.



Implementation Summary

The two-way anova mainly talks about the following:

- Is the gene expression affected by gender?
- Is the gene expression affected by smoking status?
- How the gender responds to smoking?

Answer to the above questions are evaluated by computation of F-statistics for each row.

Null Hypothesis – The gene expression is independent of gender and smoking status.

Alternate Hypothesis – The gene expression is affected by gender and smoking status.

Given data composed rows and columns representing gene_ids and their expression (intensity) respectively.

F-statistics is a ration of two chi-squares. If numerator > denominator, then null hypothesis (H0) is false

Code elaboration:

- Data is loaded and processed.
- Two design matrices N and D for estimation of [males, females, smokers, non-smokers] and [male-smokers, male-nonsmoker, female-smokers, female-nonsmokers].
- Parameters for F-statistic computation is calculated i.e. rank of N and D and degree of freedoms.
- F-statistics is computed and plotted against gene_id.
- Essentialities for P-values are calculated and p-value is calculated.
- Threshold is determined which is also called f-critical is determined by looking at f-table.
- Significant genes are selected with a condition of $p < 0.05$.

LLM was used for optimization of f-statistics computation and the logic is same as explained in the class.